



Fig. 3: Winding optimisation (Source: Under the hood of Flyback SMPS by TI)

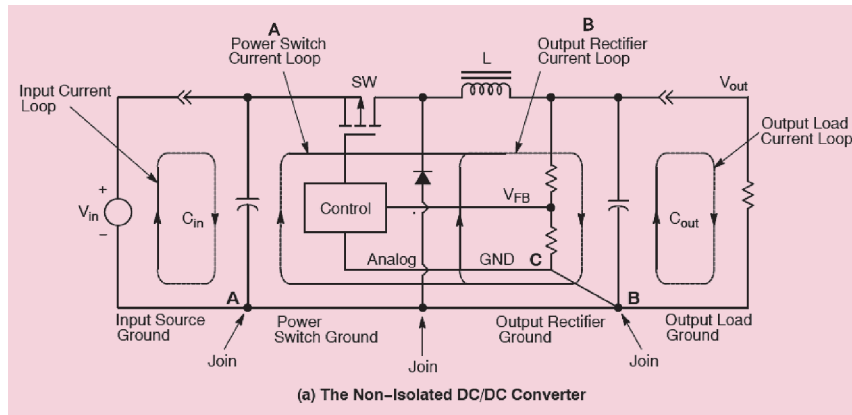


Fig. 4(a): Converter circuit (Source: SMPSRM/D, Rev 4, Apr 2014)

In diagram (c), when the primary MOSFET is turned back ON, the clamp capacitor is discharged through forward coupling using snubber winding and primary winding. In short, the energy in the form of heat is trapped and recycled back to the circuit, making the complete circuit more efficient. That's why it is known as a lossless snubber.

Optimise the magnetics

To ensure the efficient operation of SMPS, the magnetics have to be designed in a focused way. So, let us discuss it as winding placement can largely affect leakage.

We all know that the leakage inductance of primary winding should be low, but minimising the leakage inductance of low-current auxiliary winding to the primary is not a good strategy. The large leakage inductance of auxiliary winding to primary limits the energy delivered to these primary windings during commutation by increasing their CCM load range and improving their cross-regulation. The leakage should be less but for only primary winding, not for auxiliary winding. For this, we prefer to use sandwich windings.

As shown in the first block in Fig. 3, the W2 or the heavy loaded winding is sandwiched between the primary windings, which is a better

alternative to the third option where the auxiliary winding is closely packed with the W3 winding.

Another benefit of using the sandwich winding is that the auxiliary windings will be more coupled to the secondary windings, giving a tighter control for the system to optimise the output voltage sensing.

Various other factors that need to be taken care of during magnetics are:

1. Try to keep your main winding fully loaded.
2. The winding of the output with the widest load range (usually the regulated output) should have the best coupling to the primary, which means it should have the smallest leakage inductance to the primary. Primary should have the lowest leakage while auxiliary should not be considered to have low leakage.
3. Use multifiller (side-by-side) or Litz wire for optimum efficiency.
4. Operate the secondary auxiliary outputs close to the boundary between CCM and DCM. This ensures that enough energy but not too much is delivered to each. One way to accomplish this is by adding series impedance or by adding load current at minimum load.

5. When secondary windings share the same ground and polarity, use an AC or DC stack. Either stack them just after the winding for AC stack or

stack them after the diode to make DC stack. This would also improve cross-regulation.

6. Spread the winding over the complete bobbin.

Solve EMI/EMC issues

As many electronics products work at proximity at different frequencies, a lot of interference occurs. It is to be ensured that none of these devices harm us or each other, for which the products should be EMI/EMC compliant.

Since the first thought for resolving EMI/EMC issues is magnetics, certain points need to be kept in mind:

1. Always lay down the primary winding layer first, which starts at the MOSFET drain. The outer windings will act as a shield to the high switched noisy half of the primary winding connected to the MOSFET drain. Since the winding at the MOSFET has a very high noise while switching, because of which EMI/EMC issues occur, that's why put the windings lower so that other windings cover it and shield the noisy winding.

2. Always use a centre-gapped transformer. This is because there exists a fringing field, which causes magnetic flux leakage and leads to common-mode emission. For a centre-gapped core, the winding can act as a shield to the fringing field.

3. Use shields in transformer windings. Adding a shield between the primary and secondary can reduce the parasitic capacitance. The shield can be added by winding an additional layer of wire covering the whole bobbin's width. One end of this winding is left floating while the other end is connected to the primary ground.

4. The use of the transformer flux band can further reduce the EMI generated by the transformer's stray magnetic field. This flux band is grounded to the MOSFET ground through a small wire.

5. Use snubbers on power switches for controlling both the spike's transition rate and shape, and also the peak voltage. A high di/dt loop creates noise due to radiated magnetic fields. So, ensure